

The Separation-Individuation Inventory

Association with Borderline Phenomena

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The Separation Individuation Inventory (SII; Christenson and Wilson, *J. Nerv. Ment. Dis.*, 173:561-565, 1985) was designed to assess adult psychopathology construed in terms of the concepts of separation and individuation (Mahler, Pine, and Bergman, *The Psychological Birth of the Human Infant: Symbiosis and Individuation*, 1975). Its reported use is confined to the original authors' paper, which compared scores of 20 subjects with borderline personality disorder with the scores of staff control subjects. We report the association of the SII with two self-report measures of borderline personality diagnosis and phenomenology. Our British samples, from the staff and residents in a therapeutic community for severely personality-disordered clients, thus complement the original American work. The SII was found to have good internal consistency, respectable screening parameters, and high correlations with indications of trait psychopathology. We suggest a slight modification to the scoring of the inventory and two main areas for additional research.

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Theoretical Background

Mahler and her colleagues (1975), working from the drive/structure model of Freud as modified by Hartmann (1964), postulated that a process of separation and individuation is central in individual development. This process derives its name from the belief that the development of a sense of self (individuation) takes place through a "psychological birth," a separation analogous to the physical separation of birth and the cutting of the umbilical cord. Mahler et al. (1975) described four subphases of the separation-individuation process after the infant has passed through earlier phases of normal autism and a normal symbiotic phase. The first subphase, differentiation, extends from about 4 or 5 months to around 10 months. In it, the infant starts scanning the external world and showing coordinated movements away from the mother. The practicing subphase, which extends into the second year, is second, and is marked by the child's increasing absorption and triumph in both walking and speech. This is said to end in a collapse of the infantile omnipotence as the child starts to realize how small and vulnerable he or she is. That experience forms the rapprochement crisis, which represents the start of the third subphase, rapprochement, in which the child establishes a new "ambivalent" relationship with the mother. During this period, language develops rapidly, along with "reality testing." This leads into a final subphase of consolida-

tion of individuality and the beginnings of emotional object constancy. Mahler and colleagues (1975) emphasized that this is not a phase with a definable upper limit, but the start of an active process that has to be maintained throughout adult psychological life.

Psychoanalytic writers have commented on the association between disturbances in the separation-individuation process and a range of adult psychopathologies. In particular, clinical observations and anecdotal descriptions have emphasized the association between adult abnormal separation-individuation experiences and borderline personality organization (Kernberg, 1980; Zetzel, 1971). Borderline, as an adjective, is used in various contexts to describe: a state; a personality organization; a character disorder or personality diagnosis; a syndrome; and a variant of schizophrenia (Conte et al., 1980). Many aspects of these different descriptions of adult traits and states have clear analogies with Mahlerian childhood observations: In particular, the "frantic attempts to avoid real or imagined abandonment" (American Psychiatric Association [APA], 1987, p. 347) is strongly reminiscent of Mahler's descriptions of the rapprochement crisis.

The importance of separation-individuation issues in working with borderline patients in therapeutic community settings has been described by several workers (Crafoord, 1977; Producers, 1987). Particular emphasis has been placed upon attachment and dependence as symbolized in the leaving process (Wilson, 1985). Yet, despite the very considerable theoretical significance of work on borderline personality organization and on separation-individuation (inter alia Pine, 1979), there has been little empirical work on their association.

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In 1985, Christenson and Wilson reported on a 39-item questionnaire (the Separation-Individuation Inventory [SII]) designed to assess adult manifestations of disturbances in this process. Mahler's concepts of separation-individuation were essentially empirical, arising from child observation work based on psychoanalytically informed interpretations of direct behavioral observations. The SII represents an attempt to extend the empirical method into a quantification of the adult experiences of separation-individuation difficulties possibly resulting from similar difficulties in childhood, as described by Mahler et al. (1975). Christenson and Wilson (1985) showed that the SII successfully discriminated between 20 patients who met DSM-III criteria for borderline personality disorder and a normal population control group. Subsequently, there have been no reports of the use of this measure (R. M. Christenson, personal communication, 1990; psychiatric literature), perhaps because Christenson and Wilson (1985) themselves noted that "more research is needed to establish the inventory's validity and reliability and to better define its scoring."

The current study was designed to assess and quantify the association of separation-individuation phenomena with borderline personality characteristics and borderline personality disorder diagnosis in adult patients. Our expectation was that significant positive correlations would be found between SII measures and borderline characteristics. We also aimed to investigate psychometric properties of the SII in a British sample.

Setting of the Research in a Psychotherapeutic Context

This research was conducted in Henderson Hospital, a therapeutic community (TC) specializing in work with adults with severe personality disorders. The TC approach combines sociotherapy and psychotherapy in a residential setting (Dolan and Norton, 1990; Rappoport, 1960; Whiteley, 1980). The structured schedule of the community involves group work ranging from focused task-centered groups to more psychodynamic therapy groups. No individual therapy or psychotropic medication is used. Active participation of residents in their own treatment is encouraged and interactions of any type within the sociotherapeutic milieu are open for exploration and discussion. In keeping with the philosophy of the TC, research is introduced to the community as a group project within the daily program. All residents and staff on the unit participate in the group and it is seen as offering not only a forum for research, but also an opportunity for the community to explore some of the issues raised by both the "measuring" of members feelings and by the actual content of the questionnaires employed.

Methods

Subjects

Subjects were 24 patients (known as residents), 12 women and 12 men, with an average age of 27 years ($SD = 5.5$ years). Length of the current admission varied between 3 weeks and 10 months. The control group consisted of 10 staff from the same institution whose mean age was 33 years ($SD = 4.1$ years).

Procedure

Questionnaires were completed anonymously in a community group and returned immediately. All members of the community present on the day participated in the group and completed the questionnaires.

Study Instruments

Separation and Individuation Inventory. This 39-item inventory assesses adult manifestations of disturbances in the separation and individuation process. The questions cover aspects of differentiation, splitting and relationships, particularly lack of boundary between self and others, intolerance of aloneness, problems in relationship formation, and problems in interactions with others. Scores of 190 or over indicate probable pathology (Christenson and Wilson, 1985).

Borderline Syndrome Index. This is a 52-item self-report questionnaire that records the characteristics of borderline personality disorder (Gunderson, 1983) and borderline personality organization (Conte et al., 1980; Kernberg, 1984). The clinical validity of the Borderline Syndrome Index (BSI) has been shown, as has its ability to discriminate between borderline, psychotic, and normal groups (Edell, 1984). Scores of 25 and over indicate probable borderline personality disorder.

Personality Disorder Questionnaire. The Personality Disorder Questionnaire-Revised (PDQ-R; Hyler and Reider, 1987) is a 162-item self-report measure that provides diagnostic criteria for those personality disorders described in the DSM-III-R (APA, 1987). The PDQ has been shown to have test-retest reliability and compares favorably with the interrater reliability for both a DSM-III-oriented clinical interview and a semistructured interview. The specificity and sensitivity of the PDQ in borderline personality disorder diagnosis are highly satisfactory (Hurt et al., 1984).

Results

Internal Reliability of the Measures

The alpha-coefficient of internal reliability for the 39-item SII was .94 for the total sample, .91 for the residents alone, and .83 for the staff alone. For the 52-item

BSI, the corresponding values were .97, .93, and .78.² These results indicate very good internal reliability.

Description of the Sample: Prevalence of Personality Disorders, Comorbidity, and Mean BSI and SII Scores

The Personality Disorders Questionnaire indicated a high rate of personality disorder symptomatology among the residents. Eighty-three percent (95% confidence interval [c.i.] = 63% to 95%) met criteria for borderline personality disorder, and 96% (95% c.i. = 78% to 100%) met criteria for at least one cluster B diagnosis (narcissistic, histrionic, borderline, or antisocial personality disorder). These very high prevalence rates conform to clinical impressions. There was a very high comorbidity for PDQ diagnoses; three people each met criteria for 10 of the 11 diagnostic categories and only one person met criteria for one personality disorder alone. The mean number of diagnoses per resident was 6.2.

The mean scores of residents and staff were markedly different for both the BSI and the SII. The mean BSI score for residents was 30.9 (c.i. = 26.3 to 35.5), whereas that of the staff was 3.8 (c.i. = 1.2 to 6.5; $df = 32$; $t = 7.62$; $p < .0001$). The residents' mean SII score was also significantly higher than that of the staff (resident $\bar{X} = 198.7$ and c.i. = 173 to 225 vs. staff $\bar{X} = 90.8$ and c.i. = 70 to 111; $df = 32$; $t = 5.28$; $p < .001$).

Relationship Between SII and both BSI Scores and PDQ Diagnoses

The two scales were very strongly positively correlated, which indicates that although separation-individuation pathology and borderline pathology may be conceptually distinct, they are frequently linked in clinical presentations (Pearson correlation across all 34 respondents: .78; 95% c.i. = .60 to .89).

The SII scores of those meeting the PDQ criteria for each of the individual personality disorders were compared with the scores of the remainder of the sample who did not meet the criteria using independent t -tests. The differences were statistically significant for all but dependent personality disorder, with the diagnostic group having higher mean SII scores than those not having the diagnosis (Table 1). The screening parameters of the SII in relation to the PDQ diagnosis of borderline personality disorder and scores above cutoff on the BSI are shown in Table 2.

Discussion

Empirical Considerations

The study investigated the correlations between SII scores and measures of adult personality problems. These correlations were explored in a sample that contained a group expected to have very high levels of borderline pathology (residents) and another group expected to have much lower levels (staff). Our results showed very strong relationships between SII scores and diagnoses of DSM-III-R personality disorders based on the PDQ and with ratings of borderline personality organization as measured with the BSI. This relationship extends the findings of Christenson and Wilson (1985), who showed much higher SII scores in patients meeting DSM-III diagnoses of borderline personality disorder than in normal controls. Their borderline personality disorder group included 19 inpatients and two outpatients and had a mean SII score ($\pm$ SD) of 201 ± 66 (95% c.i. = 170 to 232); these results are very similar to those for our entire resident group (199 ± 62 , 95% c.i. = 173 to 225). Their control group had significantly higher scores than our staff group (121 ± 40 [95% c.i. = 115 to 127] compared with 91 ± 29 [95% c.i. = 70 to 111]; 95 c.i. difference = 5 to 55). However, their control group was composed of mailed responses from 36% of the university employees contacted, so these differences might relate to a selection bias operating in their sample, or to substantive differences in culture or specific employment group. Although the resident group differed highly significantly from the staff group on the SII, the question arises of whether this difference is specific to one or to a few of the personality disorder diagnoses or whether it is related to patient status per se. However, none of the staff met the PDQ criteria for any personality disorders and the majority of the residents met criteria for more than one PDQ diagnosis ($\bar{X} = 6.2$). This comorbidity makes it difficult to ascertain whether high scores on the SII have any specific relationship with particular personality disorders. Mahler's metapsychological model predicts a general association between separation-individuation difficulties and personality disorder.

We checked the psychometric properties of the SII by measuring the internal consistency of the items, which was high. However, the separation-individuation process was seen by Mahler and her co-workers as a multilayered and multidimensional process, so internal structure analysis of the questionnaire in larger samples might reveal some subscale structure reflecting the complexity of the process of separation-individuation.

Christenson and Wilson (1985) considered that the SII could be used as a screening instrument to separate those with separation-individuation pathology from those without it. For this purpose, they recommended

²Use of the Spearman-Brown formula to "correct" for scale length gave essentially identical results for both the BSI and SII, indicating that the slightly higher alpha-values for the BSI are a function of the greater scale length rather than of higher interitem correlations.

TABLE 1
Relationship between SII Scores and PDQ Diagnoses^a

Personality Disorder	PD Diagnosis (N)	Ss with PD ^b	Ss without PD ^b	<i>t</i>	<i>p</i>
Borderline	21	205.8 ± 58.7	104.3 ± 45.5	5.30	<.0005
Obsessive-Compulsive	16	203.1 ± 68.7	134.8 ± 62.4	3.04	.005
Avoidant	15	194.1 ± 52.2	145.5 ± 81.1	2.01	.05
Paranoid	15	203.9 ± 43.7	137.8 ± 77.7	2.89	.007
Histrionic	15	198.0 ± 66.0	142.5 ± 70.6	2.34	.025
Schizotypal	13	203.4 ± 47.1	144.4 ± 78.1	2.45	.02
Antisocial	12	214.8 ± 72.0	140.9 ± 60.6	3.20	.003
Narcissistic	12	220.8 ± 66.7	137.6 ± 59.2	3.75	.001
Dependent	11	195.0 ± 44.6	153.6 ± 81.0	1.57	.13
Schizoid	10	213.4 ± 54.2	147.6 ± 72.1	2.59	.014
Passive-Aggressive	9	223.7 ± 74.5	146.6 ± 62.2	3.03	.005

^aComparison of subjects with and without individual PD diagnoses. Independent *t*-tests, total sample *N* = 34, *df* = 32.

^b $\bar{X} \pm SD$.

TABLE 2
Sensitivity, Specificity, and Predictive Value of SII Related to PDQ and BSI Borderline Categories

	PDQ		BSI	
	-ve	+ve	-ve	+ve
SII -ve (<190)	12	9	14	7
SII +ve (>189)	1	12	1	12
Screening Parameters				
Sensitivity		57%		63%
Specificity		92%		93%
Positive predictive value		92%		92%
Negative predictive value		57%		67%

a cutoff point of 190 (total scores can range from 0 to 390). In their sample, this gave high screening specificity, so that there would be false-positive cases. This gave a sensitivity of 70%, *i.e.*, only 70% of "cases" on the basis of a DSM-III diagnosis of borderline personality disorder were detected. Our findings in this initial replication were very similar in comparing the SII against both the PDQ and BSI cutoff criteria and support this cutoff point for screening purposes.

Integration into the Work of the Therapeutic Community

The general principle of exploration of issues of personality with formal methods proved very acceptable to community members. Although participation was voluntary, all group members returned their questionnaires completed. It seemed important that the research was presented in a setting in keeping with the community philosophy. Open questioning of the research aims and discussion of feelings about "being researched" followed the exercise. The discussion in the group revealed a strong dislike for the Likert scoring method, which induced time-consuming uncertainty in the more obsessional respondents. This finding might otherwise not have emerged where there is

a more traditional separation between researcher and subject.

The involvement of both staff and residents, which creates a very unusual sampling frame, reflects an important ideological theme of the therapeutic community: communalism (Rappoport, 1960). Staff have often been used as control groups in psychological and other studies of patient groups, but it is rare for both to complete the instrument in one combined sitting and unusual for feelings and thoughts about the process then to be treated seriously as psychodynamic material that feeds back into the continuous exploration of the needs to "parent" and to be "parented."

Conclusions

Our survey shows that the SII has very high internal consistency, respectable screening parameters, and high correlations both with the Borderline Syndrome Index and with all diagnostic scales from the Personality Diagnostic Questionnaire. The work also showed that it is possible for a TC to incorporate into the therapeutic program the pilot work that is necessary for conducting any theoretically sound self-review.

The main practical problem with the use of the SII was the 10-level Likert response scale. We would strongly recommend that workers intending to use the SII consider using a visual analogue scale or a shortened Likert form of the items.

The main theoretical questions about the SII that remain to be explored are the issues of dimensionality and of the relationship to specific personality disorders. Factor analytic (or similar) studies in larger samples are required to see whether there is any consistent subscale structure to the instrument, and studies of samples with much less personality morbidity and comorbidity are needed to reveal any relationship with specific personality disorders.

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